

ACCEPTANCE TEST PROCEDURE FOR 10 KVA ONLINE UPS

1. Supply, Installation and Commissioning of 10 KVA online UPS at ADRDE will be done by vendor /Supplier.

2. After installation & commissioning, visual and technical inspection including load test will be carried out as specified below by rep ADRDE

S.No	Parameters/attribute	Value/Range	ATP Method
1.	Rating in KVA	10 KVA	Load test
2.	Type of UPS	Online UPS	By Function test
3.	Switching Technology	IGBT-PWM	Data sheet
4.	Input Voltage	AC 3-phase 300V- 450V, sine wave, 50hz	By Function test/Data sheet
5.	Output Voltage	AC Single phase 230+/- 1%, sine wave, 50 hz	By Function test/Data sheet
6.	Warranty for UPS	3 Year on site	By certificate
7.	Batteries provided with UPS	Shall be provided along with Battery	By visual inspection
8.	Type of Battery	SMF VRLA Conforming to JISC:8702, stationary lead acid Batteries conforming to IS :13369:1992 latest	By certificate/data sheet
9.	Backup time	120 minute or more	By function test
10.	Battery capacity	65 AH or more	By function test/data sheet
11.	Number of Batteries	40 or as per backup requirements	By physical inspection
12.	Warranty of battery	2 year or more	By certificate
13.	Inbuilt isolation transformer	Shall be integrated with UPS	By visual inspection
14.	Overall Efficiency %	>=90% with inbuilt isolation transformer	By data sheet
15.	Power Factor load supported	0.8 or better	By data sheet
16.	Overload limit	20% overload for minimum 10 minute	By function test/data sheet
17.	Total harmonic distortion THD	3 % Maximum @ 100 % liner load	By data sheet
18.	Other functional feature	Crest factor on full non liner load is not less than 3:1 Noise level <70dB, APFC >= 0.92 , transit repose time < 20 milliseconds ,	By data sheet/certificates

		UPS input power factor better than 0.95	
19.	Accessories	Rack of batteries, cabling 6 meter(3 meter input and output up to 3 meter) paralleling kit for synchronizing	By visual/Physical inspection
20.	Protection	-Under voltage/Over voltage at battery terminal, - Under voltage/Over voltage , - Short circuit & overload at UPS output terminal, - Over temperature protection	By Function test/visual/Physical inspection
21.	Metering	Input voltage, output voltage, input current, output current, input frequency, output frequency, input power factor, battery voltage and current.	By Function test/visual/Physical inspection
22.	Alarm and indicators	Indicator for main presence, battery charging, low battery with audible alarm, output overload with audible alarm, alarm for low battery	By Function test/visual/Physical inspection
23.	Type of design / construction	Floor standing	By visual/ Physical inspection
24.	Enclosure	minimum 1.2 mm thickness of MS sheet duty power coated	By visual/ Physical inspection
25.	Degree of protection of UPS	IP21	By certificate/data sheet
26.	DC bus voltage (in volts)	240 DC volt	By Function test/visual/Physical inspection
27.	Other constructional features	Main ON/OFF MCB, battery ON/OFF MCB , Battery fuse, inverter reset push button, static by pass, manual by pass, expandable rack facility,	By Function test/visual/Physical inspection
28.	Operating temperature	0 °C To 50 °C	By certificate/data sheet
29.	Relative humidity	95% (non condensing at 40 °C)	By certificate/data sheet
30.	Environmental test report	IS:9000, damp heat test	By certificate/data sheet
31.	Battery test report	Type test report as per govt lab/ NABL accredited lab	By certificate/test report

32.	Other requirements	UPS shall be ROHS Compliance	By certificate/test report
33.	Report and Certification	Vendor shall provide environmental test report from NABL accredited lab for UPS and all certificates mentioned in these specifications	By physical verification
34.	Make in India	Yes	By certificate/Data sheet
35.	BIS CRS certification	Shall be provided by vendor	By certificate/Data sheet

3. Battery test report and BIS, CRS/ ROHS certificate will be verified by ADRDE Rep.

Sc 'F'
(Binnami Singh)
G.D.(PARA 17)

TO 'B'
(Hari Karan Dwivedi)
Initiating Officer

TECHNICAL SPECIFICATION FOR 10 KVA ONLINE UPS

S. No	Parameters/attribute	Value/Range
1.	Rating in KVA	10 KVA
2.	Type of UPS	Online UPS
3.	Switching Technology	IGBT-PWM
4.	Input Voltage	AC 3-phase 300V- 450V, sine wave, 50hz
5.	Output Voltage	AC Single phase 230+/- 1%, sine wave, 50 hz
6.	Warranty for UPS	3 Year on site
7.	Battery provided with UPS	Shall be provided along with Battery
8.	Type of Battery	SMF VRLA Conforming to JISC:8702, stationary lead acid Batteries conforming to IS :13369:1992 latest
9.	Backup time	120 minute or more
10.	Battery capacity	65 AH or more
11.	Number of Batteries	40 or as per backup requirements
12.	Warranty of battery	2 year or more
13.	Installation and commissioning	Shall be done by vendor at ADRDE
14.	Preventive maintenance & services	Shall be provided Quarterly and as & when required during warranty period
15.	Inbuilt isolation transformer	Shall be integrated with UPS
16.	Overall Efficiency %	>=90% with inbuilt isolation transformer
17.	Power Factor load supported	0.8 or better
18.	Overload limit	20% overload for minimum 10 minute
19.	Total harmonic distortion THD	3 % Maximum @ 100 % liner load
20.	Other functional feature	Crest factor on fill non liner load is not less than 3:1 Noise level <70dB, APFC >= 0.92 , transit repose time < 20 milliseconds , UPS input power factor better than 0.95
21.	Accessories	Rack of batteries, cabling 6 meter(3 meter input and output up to 3 meter) paralleling kit for synchronizing
22.	Protection	- Under voltage/Over voltage at battery terminal, - Under voltage/Over voltage ,

		- Short circuit & overload at UPS output terminal, - Over temperature protection
23.	Metering	Input voltage, output voltage, input current, output current, input frequency, output frequency, input power factor, battery voltage and current.
24.	Alarm and indicators	Indicator for main presence, battery charging, low battery with audible alarm, output overload with audible alarm, alarm for low battery
25.	Type of design / construction	Floor standing
26.	Enclosure	minimum 1.2 mm thickness of MS sheet duty power coated
27.	Degree of protection of UPS	IP21
28.	DC bus voltage (in volts)	240 DC volt
29.	Other constructional features	Main ON/OFF MCB, battery ON/OFF MCB , Battery fuse, inverter reset push button, static by pass, manual by pass, expandable rack facility,
30.	Operating temperature	0 °C To 50 °C
31.	Relative humidity	95% (non condensing at 40 °C)
32.	Environmental test report	IS:9000, damp heat test
33.	Battery test report	Type test report as per govt lab/ NABL accredited lab
34.	Other requirements	UPS shall be ROHS Compliance
35.	Report and Certification	Vendor shall provide environmental test report from NABL accredited lab for UPS and all certificates mentioned in these specifications
36.	Make in India	Yes
37.	BIS CRS certification	Shall be provided by vendor

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